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Inclusión de espacios $\mathcal{L}^p$: caso general

Proposición 1. *Sean* $1 \leqslant p < q < r \leqslant \infty$*, entonces*

$$\mathcal{L}^p \cap \mathcal{L}^r \subset \mathcal{L}^q \subset \mathcal{L}^p + \mathcal{L}^r. \tag{1}$$

Además, como $\frac{1}{r} < \frac{1}{q} < \frac{1}{p}$*, existe* $\theta \in (0,1)$ *tal que* $\frac{1}{q} = \frac{1-\theta}{r} + \frac{\theta}{p}$ *y se cumple que*

$$\forall f \in \mathcal{L}^p \cap \mathcal{L}^r : \|f\|_q \leqslant \|f\|_p^{\theta} \cdot \|f\|_r^{1-\theta}. \tag{2}$$

***Demostración*:** Veamos primero la parte derecha de (**??**). Sea $f \in \mathcal{L}^q$, entonces

$$f = g + h \quad \text{con} \quad g(x) = f(x)\mathbb{1}_{\{|f(x)|>1\}}(x) \quad \wedge \quad h(x) = f(x)\mathbb{1}_{\{|f(x)|\leqslant 1\}}(x).$$

- Si $|f(x)| > 1$, tenemos que $|g(x)|^p = |f(x)|^p = |f(x)|^q \cdot |f(x)|^{p-q} \leqslant |f(x)|^q$ porque $p - q < 0$. Entonces, $\|g\|_p \leqslant \|f\|_q < \infty \implies g \in \mathcal{L}^p$.
- Si $|f(x)| \leqslant 1$, tenemos que $|h(x)|^r = |f(x)|^r = |f(x)|^q \cdot |f(x)|^{r-q} \leqslant |f(x)|^q$ porque $r - q > 0$. Entonces, $\|h\|_r \leqslant \|f\|_q < \infty \implies h \in \mathcal{L}^r$.

La parte izquierda de (**??**) se deduce de (**??**), que probamos a continuación. Sea $f \in \mathcal{L}^p \cap \mathcal{L}^r$.

- Si $r < \infty$, por la desigualdad de Hölder con $p_1 = \frac{r}{(1-\theta)q}$ y $q_1 = \frac{p}{\theta q}$ exponentes conjugados, tenemos que

$$\begin{aligned}\|f\|_q^q &= \int_X |f|^q \,\mathrm{d}\mu = \int_X |f|^{(1-\theta)q} \cdot |f|^{\theta q} \,\mathrm{d}\mu \\ &\overset{\substack{\text{Hölder} \\ \downarrow}}{\leqslant} \left(\int_X |f|^{p_1(1-\theta)q} \,\mathrm{d}\mu\right)^{1/p_1} \cdot \left(\int_X |f|^{q_1 \theta q} \,\mathrm{d}\mu\right)^{1/q_1} = \\ &\leqslant \left(\int_X |f|^r \,\mathrm{d}\mu\right)^{(1-\theta)q/r} \cdot \left(\int_X |f|^p \,\mathrm{d}\mu\right)^{\theta q/p} = \|f\|_r^{(1-\theta)q} \cdot \|f\|_p^{\theta q}.\end{aligned}$$

- Si $r = \infty$, lo dejamos como ejercicio.

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